Ridge Regression

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Date: 2025-11-17
Repository: https://github.com/aliozgur/ml-ridge-lasso

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What is Ridge Regression?

Ridge regression is a linear regression technique that adds a penalty to the sum of squared coefficients to prevent overfitting. It shrinks coefficients toward zero but never makes them exactly zero.

Mathematical Formulation

Ridge minimizes the following objective:

$$\text{RSS} + \lambda \sum_{j=1}^{p} \beta_j^2$$

- RSS: Residual Sum of Squares
- λ (lambda): Regularization parameter controlling the strength of the penalty
- β_j: Coefficient for feature j

A Type of Linear Regression with Regularization

Ridge regression is a linear regression technique that adds a penalty to the sum of squared coefficients. Unlike standard OLS, it shrinks coefficients toward zero but never makes them exactly zero.

The Problem of Overfitting and Multicollinearity

Standard linear regression can overfit when there are many features or when features are correlated (multicollinearity). Ridge regression addresses this by penalizing large coefficients, which reduces model complexity and improves generalization.

Ridge Regression Formula

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Why Ridge Regression?

- Prevents overfitting by shrinking coefficients
- Handles multicollinearity (correlated features)
- Keeps all features (unlike Lasso which can eliminate features)
- Works well when many features contribute small effects to the outcome

Simple Example of Ridge Regression

Suppose we have 3 features and 100 samples:

RSS	=	50
β_1	=	2.0
β_2	=	1.5
β_3	=	0.8
λ	=	1.0

Ridge minimizes the sum of the RSS and the penalty term:

$$\text{RSS} + \lambda \sum_{j=1}^{p} \beta_j^2$$

$$= 50 + 1.0 \times (2.0^2 + 1.5^2 + 0.8^2)$$

$$= 50 + 1.0 \times (4.0 + 2.25 + 0.64)$$

$$= 50 + 6.89$$

$$= 56.89$$

Ridge finds the coefficients that minimize this total value.

Ridge Regression Formula

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```
# Ridge Regression Formula
import numpy as np
import matplotlib.pyplot as plt

# Data
features = ['Feature 1', 'Feature 2', 'Feature 3']
coefficients = [2.0, 1.5, 0.8]

# Calculation
rss = 50
lambda = 1.0
penalty = lambda * (coefficients[0]**2 + coefficients[1]**2 + coefficients[2]**2)
total = rss + penalty

print("Ridge Regression Formula")
print("=" * 40)
print(f"RSS = {rss}")
print(f"λ = {lambda}")
print(f"Penalty = λ × (β1² + β2² + β3²)")
print(f"Penalty = {lambda} × ({coefficients[0]}² + {coefficients[1]}² + {coefficients[2]}²)")
print(f"Penalty = {lambda} × ({coefficients[0]**2} + {coefficients[1]**2} + {coefficients[2]**2})")
print(f"Penalty = {lambda} × {round(coefficients[0]**2 + coefficients[1]**2 + coefficients[2]**2, 2)}")
print(f"Penalty = {round(penalty, 2)}")
print(f"Total = {round(total, 2)}")

# Visualization
fig, axes = plt.subplots(1, 2, figure size=(12, 5))

ax1 = axes[0]
ax1.bar(features, coefficients, color=['#3b82f6', '#3b82f6', '#3b82f6'])
ax1.axhline y=0, color='gray', linestyle='-', linewidth=0.5)
ax1.ylabel('Coefficient Value')
ax1.title('Coefficients')
ax1.grid(axis='y', alpha=0.3)

ax2 = axes[1]
ax2.bar(['RSS', 'Penalty', 'Total'], [rss, penalty, total], color=['#3b82f6', '#ef4444', '#22c55e'])
ax2.ylabel('Value')
ax2.title('Ridge Regression Formula')
ax2.grid(axis='y', alpha=0.3)

plt.tight_layout()
plt.show()
```

```
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